

Element G

Element G: Construction of a Testable Prototype

Cardboard Prototype For Size

Date: March 5-6

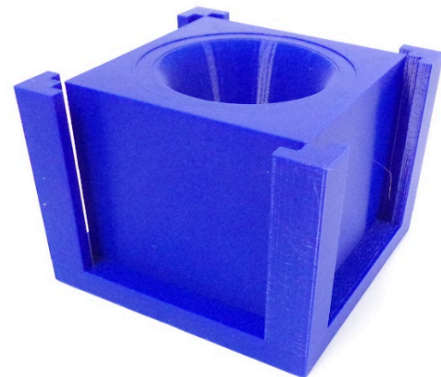
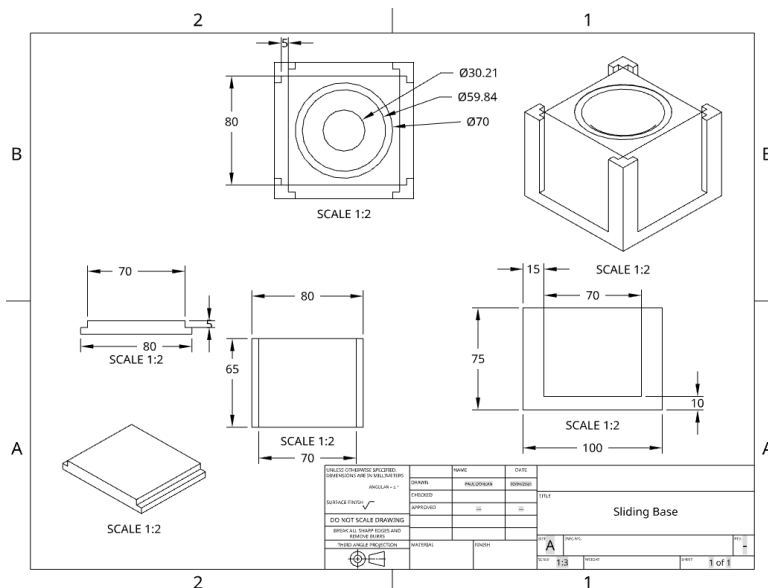
	Details	Tools and Manufacturing Methods used
Step 1	Gather cardboard to make a prototype	Cardboard
Step 2	Use a ruler and pencil to measure out sizes of panels	Ruler & Pencil
Step 3	Cut out pieces of cardboard	X-Acto Knife
Step 4	Organize pieces of cardboard to make a box	
Step 5	Glue panels together making a box	Hot Glue
Step 6	Cut out hole on top and bottom to fit microphone through	X-Acto Knife



This is our prototype made from cardboard, and this was mainly intended to get an idea of the right size for our mic flag. The pictures show the side, top, and corner of the flag. The final picture is a picture of the parts we used to make the prototype.

Prototype Construction of Design #1 - Mic Flag with Slides Date: March 13-14

	Details	Tools and Manufacturing Methods used
Step 1	Made sketch on paper	Paper and pencil
Step 2	Copied sketch over to onshape with dimensions based on cardboard flag	Onshape
Step 3	Placed 3D designs into assembly on onshape to confirm fit	
Step 4	Export the base of the design	
Step 5	Render base in Prusa to 3D print	Prusa Software
Step 6	3D Print Design	3D Printer & Filament

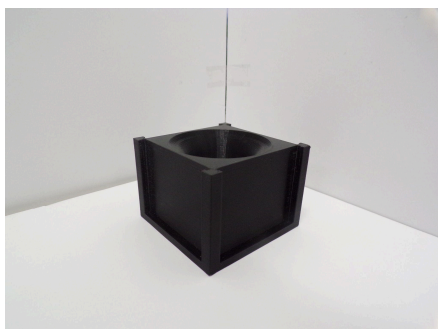


This was our first sketch & variation of our sliding mic flag. After printing it out we realized it was far too large, and with the corners only being supported at one point it would easily break off.

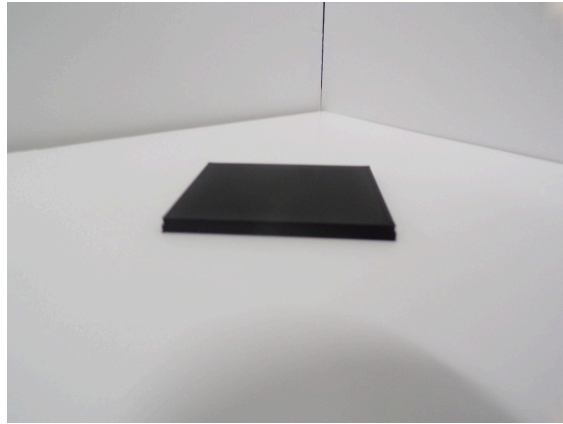
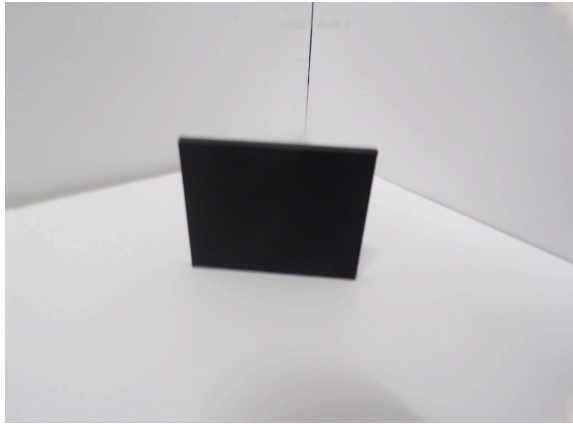


2nd Prototype Construction of Design #1 - Mic Flag with Slides Date: 3/26-3/28

	Details	Tools and Manufacturing Methods used
Step 1	Went back into onshape to make the size of the mic & panels smaller since our first design was too big.	Onshape
Step 2	Placed 3D designs into assembly on onshape to confirm fit	
Step 3	Export the base of the design & panel	
Step 4	Render base & 1 panel in Prusa to 3D print	Prusa Software
Step 5	3D Print Flag & 1 Panel	3D Printer & Filament
Step 6	Sand down first panel until it fits securely in the flag	Sand paper
Step 7	Fix size of panels in onshape again to be consistent with sanded down panel	Onshape
Step 8	Export the base of the panel	
Step 9	Render panel in Prusa to 3D print	Prusa Software
Step 10	3D Print Panel	3D Printer & Filament
Step 11	Put panels in base to make sure they fit.	



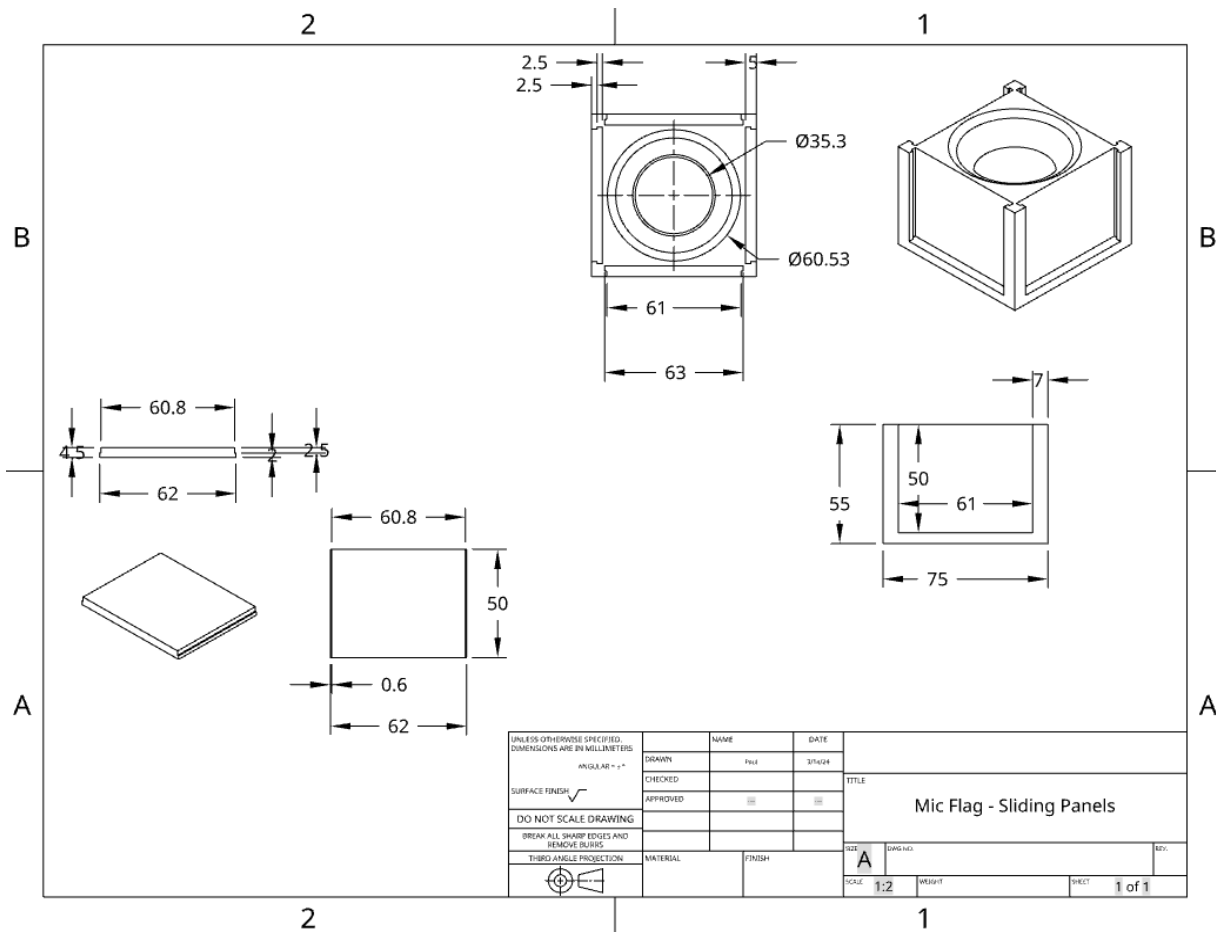
This was our original sliding base without slides, which after adjusting our last one, made it more durable and a better size compared to our first mic flag



These are our sliders that we slide into our mic flag to provide the advertising space for the logos



This is our sliding mic flag put together, and our sliders just slide into place straight into the base



CAD Drawing Of The Sliding Mic Flag

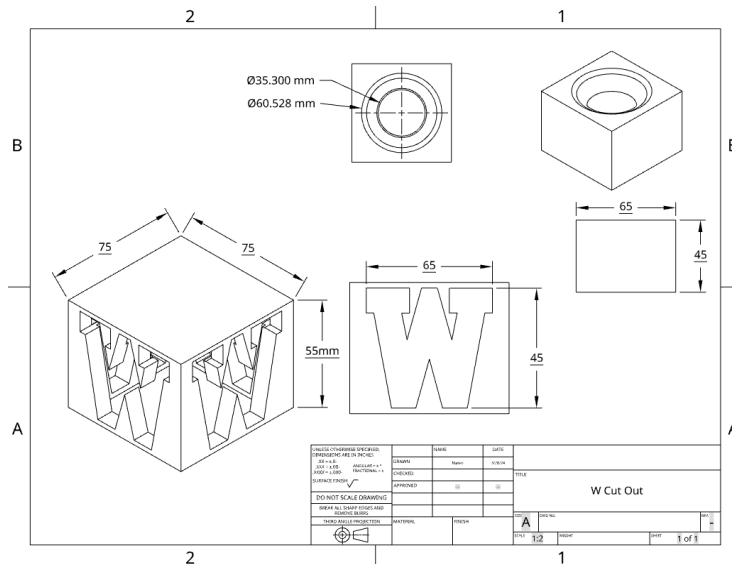


1st Prototype Construction of Design #2 - Mic Flag with W Cut Out Date: 4/1-4/3

	Details	Tools and Manufacturing Methods used
Step 1	Went into onshape to make a design that is a box and outer shell with a cut out for the W (Used picture to make cut out)	Onshape
Step 2	Placed 3D designs into assembly on onshape to confirm fit	
Step 3	Export the base & shell of the design	
Step 4	Render base and shell	Prusa Software
Step 5	3D Print Base and shell flag	3D Printer & Filament
Step 6	Sand down base until it fits inside of flag	Sand Paper
Step 7	Resize base in onshape so that it fits new dimension of the sanded down base	Onshape
Step 8	Export new base	
Step 9	Render new base	Prusa Software
Step 10	3D Print new base	3D Printer & Filament
Step 11	Glue new base to shell	Super Glue



Shell (first picture) and Base (second picture) of W mic flag. The 3rd pictures is of the two pieces being put together



This is our final W cutout mic flag & sketch. We decided to have an orange background and a blue base after talking with Mr O'Farrell and getting his opinion on what color combination was better.

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Soft foam	Increases friction on the inside of the mic flag to hold it in place	amazon	\$0.24	1	\$0.24
PLA 3D Printer filament	This was the material used for the overall structure of the mic flag	amazon	\$6.00	257g	\$24
Paper	Used to print logos on	amazon	\$0.02	1 Sheet (of 500)	\$10
Clear Tape	Used to protect logos from scratching and fading	amazon	\$0.10	4 Pieces of Tape	\$3.60
Printer Ink	Used to print the logos	amazon	\$0.20	4 50mmX60mm Rectangles	\$55

Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
The pictures on the mic flag do not fade High Priority	Test them by putting them in a car for a day or two
Size of logo must be 75mm by 50 mm High Priority	Have the flags corrected to the right size and have them fit all the holders they were designed to fit.
The mic flags should be durable High Priority	Have them made out of durable materials, so if they do fall they won't completely break or no severe damage will be down.
Mic flags should be replaceable when damaged Moderate Priority	Have different flags on standby easy replacements for when they get damaged in the field.
Mic Flag must be able to fit on multiple types of mic holders Moderate Priority	Ethier make a multi-fit flag or make different flags for different size holders.
Corners of the mic flag must not be sharp Low Priority	Use sandpaper to dull the sides so someone won't get hurt using the flag.
mass of prototype <100g Low Priority	Mic flag should not be heavy

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
Presentability (Promotion of School Logos)	How The Flag Will Look On Camera	Since the prototype has logos, we can record a video and see how well the logos show up on camera.
Reproducibility	How Easy The Flag Is To Reproduce	We want our mic flag to be easy to remake and be remade within 12 hours. Since we are testing the same manufacturing style with the prototypes as we will use for the final product, we will be able to figure out how long it will take us to make a new flag.
Durability	How Strong Is The Flag When Dropped/Hit	Our prototype has the same materials as the final design so we can test the durability of the final design with the prototype
Price	How Much Money To Make The Flag	Since the prototypes are made out of the same material as our final product, we will get an idea of price as we complete our prototype.
Replaceability	How Easy It Is To Change What Logos Are On The Flag	We can print out multiple panels and figure how long it takes to replace all 4 panels (only works on slide flag).

Addressing Non-Testable Attributes (G4)

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Long Term Durability	As time goes on, the flags will wear down and degrade. We are unable to test long term use as it would take too much time for what we have in this class and the results that we would yield is not representative of long term use.